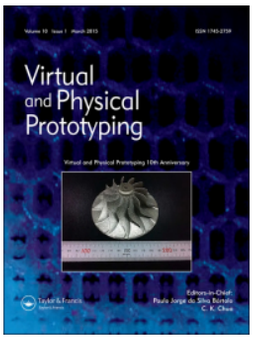

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Wire-based friction stir additive manufacturing towards isotropic high-strength-ductility Al-Mg alloys

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ABSTRACT
The formation of eutectic Al-Mg2Al3 phases along the grain boundaries makes it difficult to improve the mechanical properties of high-magnesium-content aluminum alloys. Here, wire-based friction stir additive manufacturing(W-FSAM) was proposed as a novel solid-state additive manufacturing technology. The shearing, transport, and thermo-plasticisation processes of the deposited materials were achieved by a rotational tool and a stationary barrel. The original Mg2Al3 phases were refined and dissolved into the matrix and formed supersaturated solid solution structures. The refined grains with a diameter of 1.27 ± 0.64 μm were achieved through the dynamic recrystallization process. The components achieved isotropic mechanical properties due to the homogeneous equiaxed fine-grain structure. The ultimate tensile strength reached 415 ± 11 MPa with an elongation of $31.0 \pm 2.0\%$, superior to the commercial 5xxx products. The enhanced strain hardening capacity was achieved by the suppressed emission of dislocations from grain boundaries and the interaction between the supersaturated solid solute atoms and mobile dislocation.

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KEYWORDS
Friction stir additive manufacturing; Al-Mg alloys; high magnesium contents; mechanical properties; strain hardening capacity

1. Introduction

5xxx series aluminum alloys, regarded as Al-Mg alloys, are sparked by good corrosion resistance and high mechanical performances since they can be strengthened via the solution of Mg atoms and be further strengthened by mechanical working [1–4]. During the melting and re-solidification process, Eutectic Al-Al_3Mg_2 phases are easily formed along the grain boundaries deteriorating the enhancement of mechanical properties for high-magnesium-content aluminum alloys [5–7]. As new alternatives to obtaining structures with highly complex shapes without special facilities, additive manufacturing technologies are divided into fusion-based and solid-state-based types [8, 9]. Utilising additive manufacturing technologies, specific micro-structures could be designed and obtained to achieve high-performance materials and multifunctional metamaterials for diverse materials (Inconel alloys, stainless steel, composites, etc.) [10–12]. Taking high-magnesium-content aluminum alloys (≥6 wt% [13, 14]) into consideration, fusion-based additive manufacturing technologies cause micro-cracks owing to the wide solidification range influenced by the Mg contents [15]. The major challenges for manufacturing high-magnesium-content aluminum alloy components are to avoid eutectic phases, cracks, pores, and elemental vaporisation to achieve high mechanical performances.

Solid-state-based additive manufacturing technologies can effectively avoid defects, like pores and cracks due to the absence of the melting-solidification process and additionally avoid the eutectic Al-Al_3Mg_2 phases by severe plastic deformation [16, 17]. Furthermore, they can fabricate components with homogeneous fine-grain microstructures and excellent mechanical properties, sparked by the low heat input and severe plastic deformation [18]. For high-magnesium-content aluminum alloys, equilibration of the supersaturated solid solution could be achieved by the grain boundary diffusion accelerated by fluxes of vacancies produced due to severe plastic deformation [19]. Thus, various shapes of high-magnesium-content aluminum alloy components could be directly fabricated with enhanced mechanical performance via solid-state-

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based additive manufacturing technologies. However, solid-state-based additive manufacturing technologies were mostly achieved by feeding consumable plates and rods, such as friction stir additive manufacturing (FSAM), friction surfacing (FS), and additive friction stir deposition (AFSD) [20, 21]. The FSAM was conducted by a non-consumable rotating tool introduced into the upper plate to generate heat and complete the bonding with the lower plate [22]. The FS method utilised consumable rods as feedstocks, which rotate and directly friction with the substrate to generate heat and be deposited on the surface [23]. In the AFSD method, rotating tool friction with the substrate generates heat, and rods can be delivered through the hollow tool and deposited on the substrate [24]. However, the feedstock of rods due to the limited length needs to be repeatedly performed, leading to poor bonding at the transition positions [25].

In this work, wire-based friction stir additive manufacturing (W-FSAM) was proposed as a novel solid-state-based additive manufacturing technology that can achieve continuous material deposition, benefiting the manufacturing of large-scale components. Enhanced strain hardening capacity was achieved in Al-7Mg alloy components via wire-based friction stir additive manufacturing. The strain hardening capacity and the relevant mechanisms were discussed to illustrate the high ductility. The characteristics of microstructures and the formation mechanisms of the thermo-plasticisation and deposition process were also discussed in detail.

2. Materials and methods

2.1. Materials and technical routes

Al-7Mg alloy wires with a diameter of 1.6 mm were used, whose composition was listed in Table S1. The average Mg contents of these wires reach 7.55 wt.%, which is much higher than the equilibrium solubility limit of Mg in Al at ambient temperature [26]. The W-FSAM technical routes are presented in Figure 1. W-FSAM is a novel

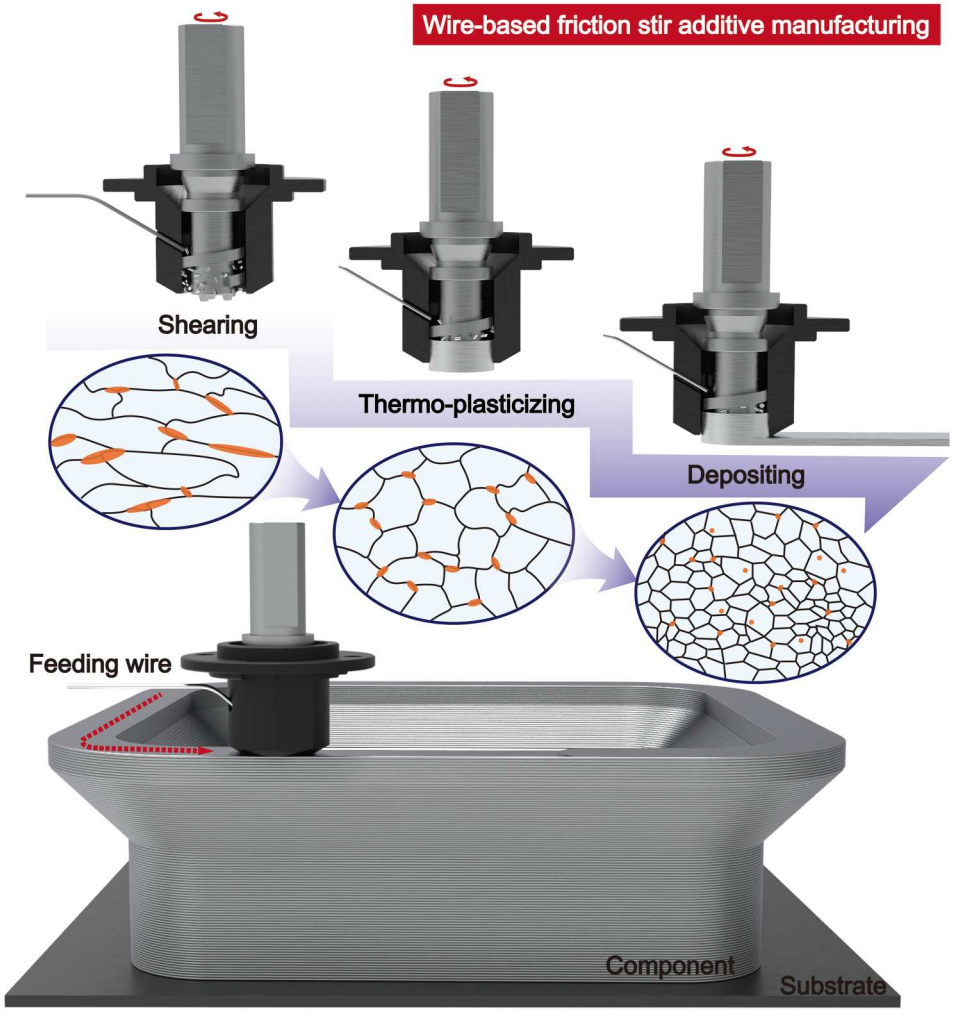


Figure 1. Schematic of the W-FSAM.

solid-state additive manufacturing technology, that introduces metal wire feedstock into the tools to be thermo-plasticised by frictional and deformation heat. Large-scale structures can be fabricated with continuous feeding of wire materials. W-FSAM technology was performed with a rotational tool and a stationary barrel fitted, providing spaces for thermo-plasticised materials. The relative rotational movement between the rotational tool and the stationary barrel could drive the fragmented particles and thermo-plasticised materials to be deposited and generate frictional/deformation heat. Severe plastic deformation was introduced as the feedstocks transferred long the screw and stirred by the pin. The grains could be refined, and the precipitations could be broken, dissolved, and re-participated. The W-FSAM process was carried out with three steps: shearing, thermo-plasticisation, and deposition. (a) Wires were cut into fragmented particles by the rotational screw during the shearing process. Fragmented particles were transported downwards along the spiral grooves by gravity and interfacial friction; (b) friction and severe plastic deformation resulted in the elevated temperature and provided the heat for the thermo-plasticisation process after the densification process; (c) the thermo-plasticised materials were finally deposited by the further transport of the spiral grooves and mixed with the substrate or the former layers by the rotational pin under the tool. The materials were densified before the thermo-plasticisation process benefiting heat production and transfer. The pressure for the densification process was provided by the continuous particles filled into the barrel. The bonding of the adjacent layers could be further strengthened by the forging process of the barrel. W-FSAM process achieved continuous material deposition and fabricated components with homogeneous properties showing great potential for manufacturing large-scale components.

The samples were fabricated via W-FSAM at a rotational velocity of 1200 rpm, a travelling speed of 600 mm/min, and a wire feeding rate of 5 kg/h. The thickness of single layers was 0.9 mm, and the length of the pin was 2.0 mm, indicating that when one layer was being deposited the former layer could be stirred and go through the dynamic recovery and dynamic recrystallization process. The rotational velocity was required to match the wire feeding rate and travelling speed to ensure the wire feedstock could be sheared into particles continuously and the thermo-plasticised materials could be stirred sufficiently.

2.2. Characterisation

The component deposited in reciprocating linear deposition paths was used for microstructural and mechanical characterisation. The composition of the deposition sample was analyzed using an X-ray fluorescence spectrometer (XRF, Vanta Element-S). Specimens used for microstructural characterisation were cut and prepared from sampling components with the size of 150 mm × 50 mm × 18 mm, as shown in Figure 2a. The specimens were etched by 10% HF reagents and Barker's reagents for optical microscopy (OM, KEYENCE VHX-1000E). The microstructural evolution of the as-deposited Al-7Mg alloy component was also characterised using a scanning electron microscope (SEM, HITACHI SU5000) equipped with EBSD. The X-ray diffractometer (XRD, PANalytical X'PERT) was utilised to identify the phase constitution using a Cu Kα ($\lambda = 0.1541$ nm). The specimens for electron backscattered diffraction (EBSD, Oxford Instrument C-Nano) were prepared by electropolishing using a solution of 90% C_2H_5OH + 10% $HClO_4$ at 20 V for 80 s at − 30 °C. Three uniaxial tensile samples in the horizontal direction and six uniaxial tensile samples in the vertical direction were taken, whose dimensions are presented in Figure 2b. The tensile tests were performed on Instron 5565 uniaxial tensile equipment with a strain rate of 1×10^{-3} s^{-1} as shown in Figure 2c.

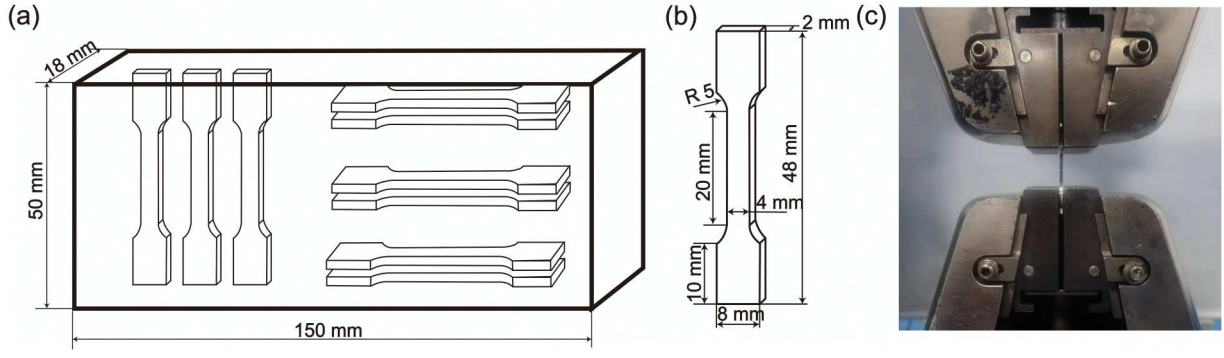


Figure 2. Configuration of the tensile tests: (a) sampling position, (b) dimensions, and (c) the fixtures.

3. Results

3.1. Formation

A rectangular component and another component used for tests have been successfully designed and manufactured as shown in Figure 3a, b. Another component with an inclined structure weighing 5 kg was also designed and fabricated at once as shown in Figure 3c. No transition positions existed due to the continuous depositing process, benefiting the homogeneous mechanical performance. Especially, the inclined structure was fabricated with a horizontal offset of 1.0 mm between adjacent layers, and the slope reached 42°. The special structure was successfully fabricated attributed to the relatively low pressure (about 6 kN) during the process and sound bonding between the adjacent layers, showing a potential to fabricate unsupportable structures. The components possessed layered structures with a thickness of 0.9 mm. Sound bonds were achieved by the stir and forging actions from the tools, benefiting the mechanical performance along the deposition direction. Compared with melt-based additive manufacturing and other solid-state-based additive manufacturing technologies [27–29], the Al-7Mg alloy components fabricated by W-FSAM performed a pretty appearance, making the components able to be applied immediately with few post-processing. The homogeneous macro morphologies indicated the stability and continuity of the manufacturing process. Therefore, this technology showed great potential for manufacturing large-scale Al-7Mg alloy components.

3.2. Microstructural analyses

The macro – and microstructural features in the cross-section of the deposited materials are shown in Figure 4. The sampling for tests was 45 mm high with an effective width of 18 mm. The thickness of a single layer was about 0.9 mm. The deposited component is dense with layered structures as shown in Figure 4a. Due to the W-FSAM being a solid-state additive manufacturing technology, it can avoid the pores and cracks due to the absence of melting-solidification. Indiscernible interfaces and sufficient materials flow were observed in magnified images between the adjacent layers, affected by the pins under the tool as shown in Figure 4b-e. Densification of multi-layers and material mixture due to the rotational pin and stationary barrel played an important role in achieving sound bonding between adjacent layers, which was beneficial for the homogeneity of the components. Equiaxed fine grain structures could be observed in Figure 4b-e, which were formed during the dynamic recovery and dynamic recrystallization process. It could enhance the strength by fine grain strengthening mechanism.

To estimate the effects of multiple thermal cycles on the homogeneity of the components, inverse pole figure (IPF) maps of deposited materials were presented and described the grain orientation distribution as shown in Figure 5a, d. The top and bottom sections were both observed and compared. The top section was selected at a 5 mm distance from the upper surface, and the bottom section was at a 5 mm distance from the undersurface. The cross sections were perpendicular to the travelling direction. Both of them were in the middle of the horizontal position. Equiaxed fine grains were observed in both two samples, indicating that the dynamic recrystallization occurred during the process. It was attributed to the thermo-mechanical coupling process by the friction stir process of the tool. The average grain size of the top section was almost the same as that of the bottom section and reached respectively 1.27 ± 0.64 μm and 1.27 ± 0.69 μm, as shown in Figure 5b, e. Therefore, the grain size was

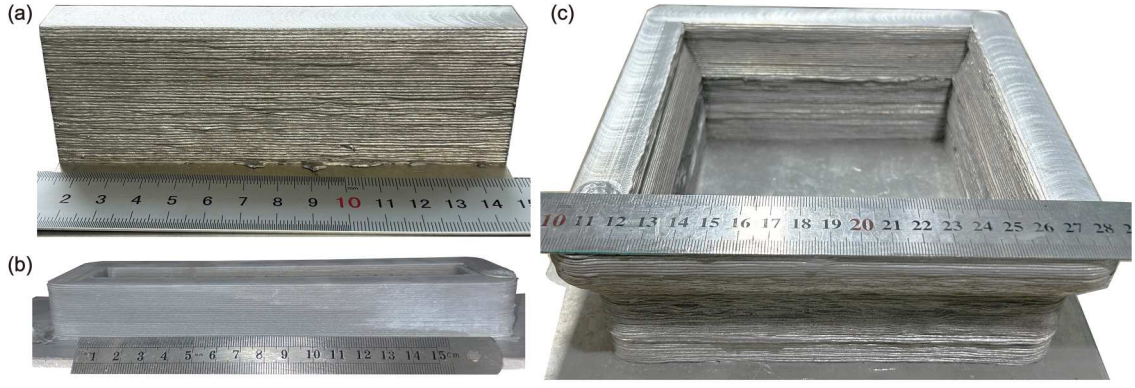


Figure 3. Macro morphology of fabricated components: (a) sampling component for tests, (b) rectangular component (c) 5 kg weight component with an inclined structure.

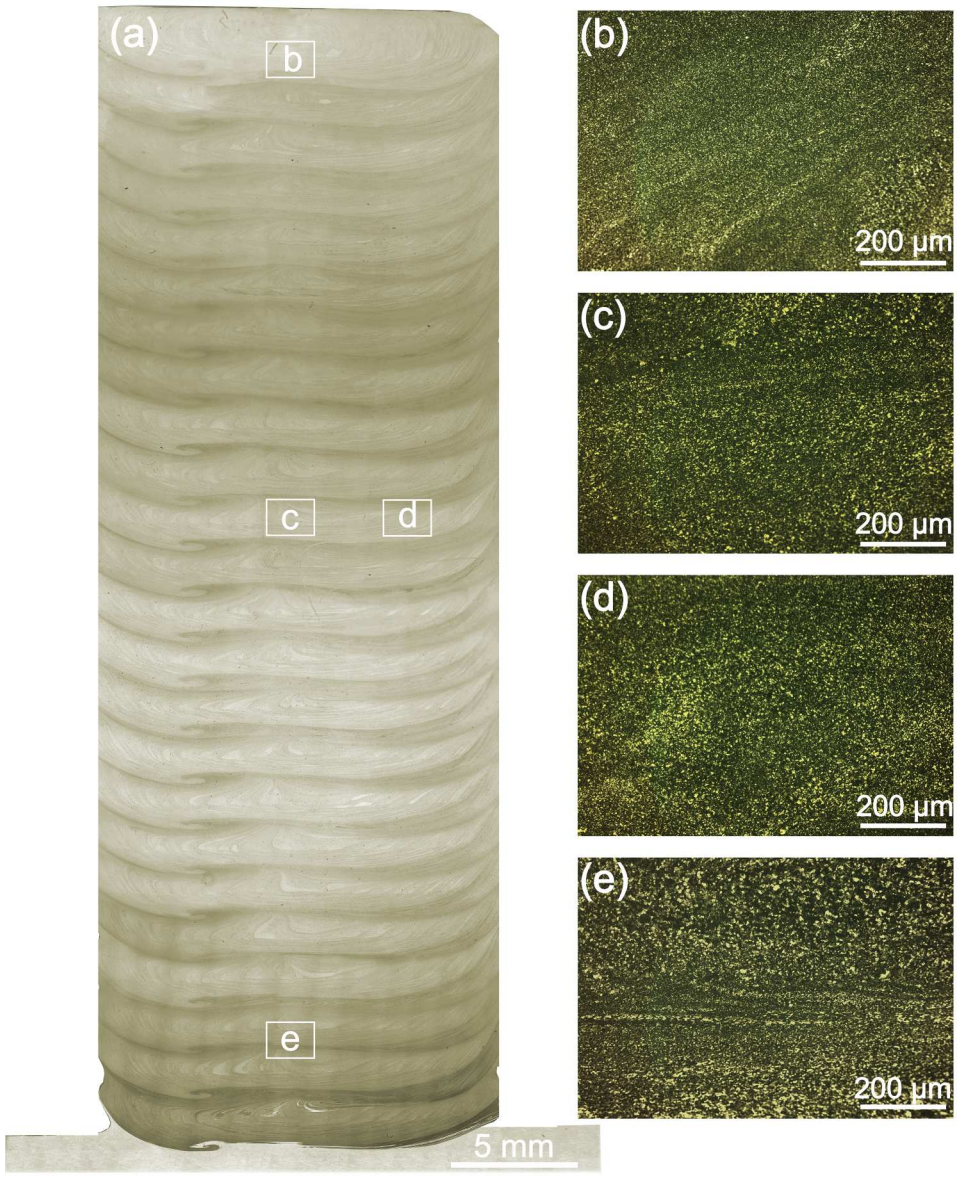


Figure 4. OM images of the deposited materials: (a) the macrostructure, (b) the microstructure of the top section, (c) the microstructure of the middle section, (d) the microstructure of the interface, and (e) the microstructure of the bottom section.

not affected by the multiple thermal cycles here. The components performed were highly homogeneous. When the Al atoms are substituted with the Mg atoms, 12% atomic misfit triggers lattice distortion, causing pinning effects of dislocations, which hindered the movement of grain boundaries, promoting the homogeneity of the component [30]. The equiaxed fine grains enhanced the mechanical properties of deposited materials according to the Hall-Petch relationship. Materials underwent severe plastic deformation during the W-FSAM process, and the geometrically necessary dislocation (GND) density reached 3.61×10^{14} m^{-2} for the top of the sample. As for the bottom of the sample, that was also estimated to be 3.68×10^{14} m^{-2}. The density of GNDs identified the homogeneity of microstructures again. Thus, the homogeneity of microstructures caused consistent mechanical properties of the components, which was beneficial for the components to be applied.

Dislocations were generated and multiplied due to the severe plastic deformation process during W-FSAM. The high density of dislocations provided the driving force for dynamic recrystallization. During the W-FSAM process, dynamic recrystallization happened in the alloys, as the grain orientation spread (GOS) maps showed in Figure 6a, d. The grain orientation spread was calculated and shown in Figure 6b and e. The key alloying element Mg resulted in a decrease in stacking fault energy, inhibiting the cross-slip and the dynamic recovery process and dislocation pile-up increased the

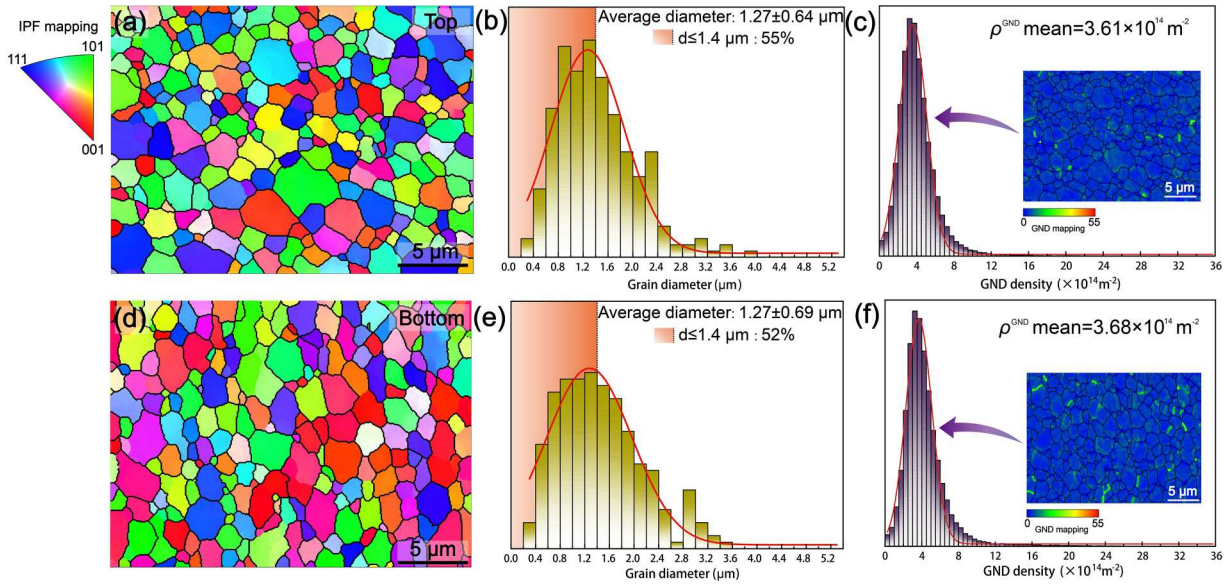


Figure 5. IPF map and analyses of different positions: (a) IPF map of top section of the depositions, (b) grain diameter of top section, (c) calculated GND density of top section, (d) IPF map of bottom section of the depositions, (e) grain diameter of bottom section, and (f) calculated GND density of bottom section.

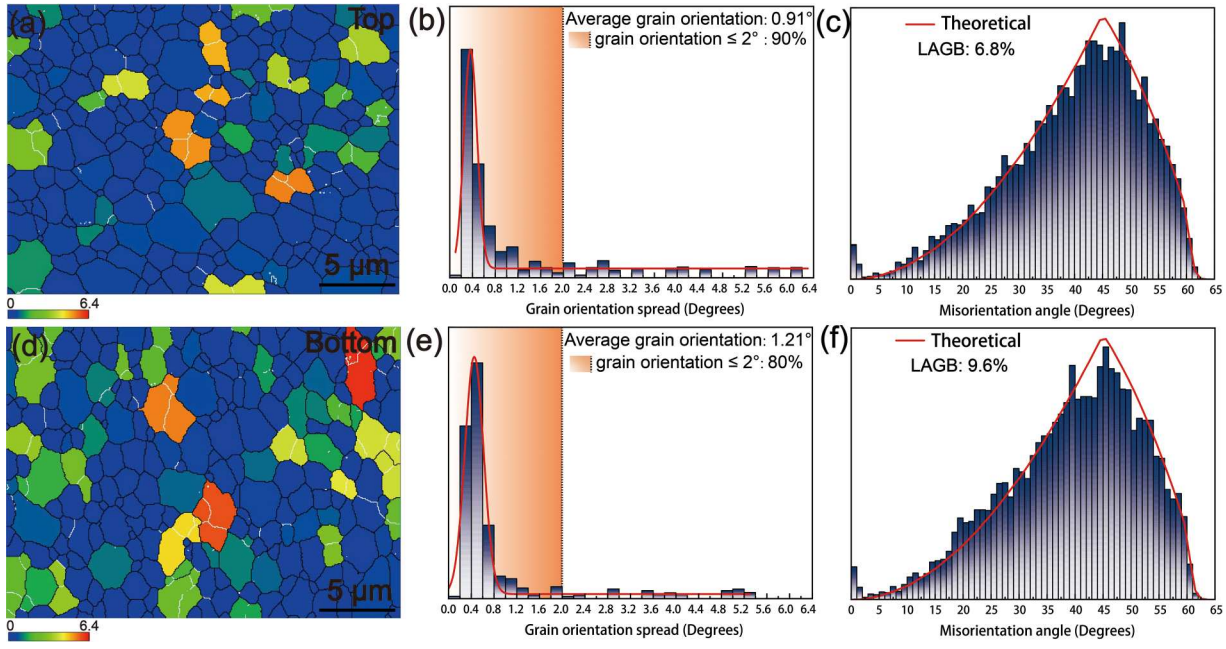


Figure 6. GOS maps and analyses of different positions: (a) GOS map of the top section of the depositions, (b) grain orientation spread from (a), (c) misorientation angle from (a), (d) GOS map of bottom section of the depositions, (e) grain orientation spread from (d), and (f) misorientation angle from (d).

stored energy and attributed to the dynamic recrystallization process [31]. In the top section, 90% of the grains had an orientation difference smaller than 2 degrees, and it was 80% for the bottom section, indicating that multiple thermal cycles contributed to the static recovery of the former depositions. Misorientation angles of the top and bottom sections were also shown in Figure 6c, f. The misorientation angle distribution generally fitted the theoretical random orientation curves, indicating that the depositions were lack of texture. The grain boundaries were divided into low-angle grain boundaries (LAGBs, 2° < misorientation angle < 15°) and high-angle grain boundaries (HAGBs, misorientation angle >15°), which were related to the dynamic recovery and recrystallization process [32]. The sub-structured grains were formed in the dynamic

recovery process, and adjacent sub-structured grains were separated by LAGBs. As for the dynamic recrystallization process, the dynamic recrystallized grains formed via the consumption of stored energy, and the angle of the grain boundaries between adjacent dynamic recrystallized grains were HAGBs. The fractions of LAGBs were 6.8% and 9.6% for the top and bottom sections, respectively. The materials went through dynamic recrystallization via thermo-mechanical treatment and more sub-structured grains were formed during the following multiple thermal cycles resulting in the fraction of LAGB increasing.

The XRD patterns of Al-7Mg wires and the W-FSAM samples are shown in Figure 7 For both samples, only the peaks of the Al phase were observed and no reflections of other crystalline Mg-containing phases were detected. A least-squares refinement was done to evaluate the lattice constants according to Figure 7a, showing the lattice constant of the wires and deposited sample were 4.0797 ± 0.0002 Å and 4.0815 ± 0.0012 Å, respectively. Both of them are larger than the lattice constant of pure Al (a = 4.0413 Å) [5], indicating that high contents of atoms were dissolved in the Al matrix. Moreover, the lattice constant of the deposited sample was slightly higher than that of wires, which suggested that through the elevated-temperature severe plastic deformation of the W-FSAM, more solute atoms dissolved into the Al matrix as shown in Figure 7c, d. A backscattered electron detector was utilised to examine the distribution of precipitates in wires and deposited layers. Compared with the wires, the samples possessed finer precipitates with dispersive distribution as pointed out in Figure 7d. It was obvious that the fraction of Mn-rich phases with lighter morphologies decreased after the W-FSAM process. Precipitates were dissolved and re-precipitated partly and the large-size precipitates could be broken up by severe plastic deformation from the pin under the tool, as for the darker phases regarded as the Mg-rich phases, a higher volume fraction of Mg-rich phases was found in wires. Original coarse phases were significantly decreased after the W-FSAM process, illustrating that Mg-rich phases were refined in this process. Homogeneously distributed precipitates could enhance yield strength via

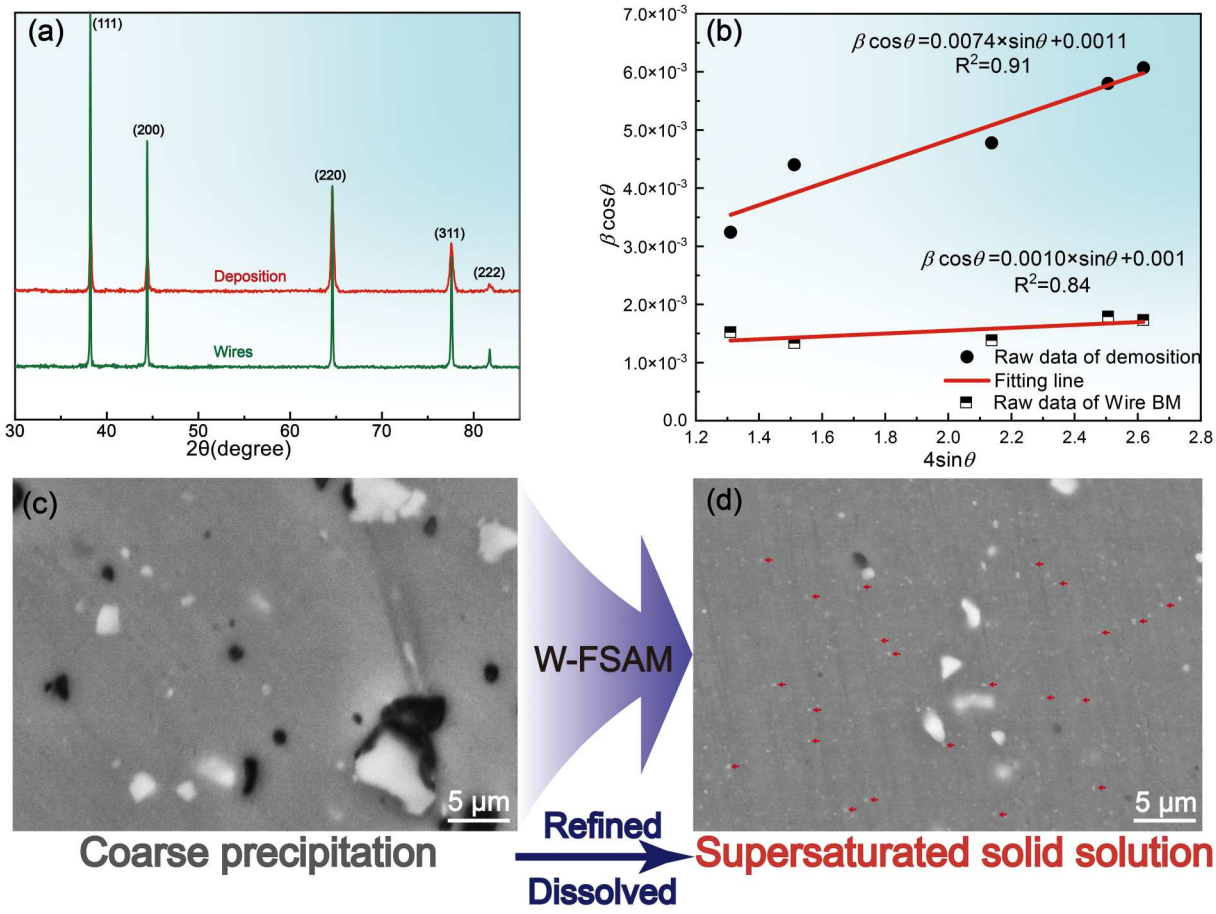


Figure 7. XRD patterns and SEM images of wires and depositions: (a) XRD results, (b) estimates of the dislocation density, (c) precipitate distribution in wires, and (d) precipitate distribution in Al-7Mg depositions.

Orowan strengthening. Based on the lattice constants, the Mg concentrations in the matrix of the deposition could probably be estimated via the following Al-Mg binary alloys empirical equation [33]:

$$C^{Mg} = \frac{\bar{a} - 4.048}{0.00452} \quad (1)$$

where C^{Mg} is the Mg concentration (at.%) in the matrix. $\bar{a}$ is the lattice constant. The Mg concentration (at.%) in the matrix was about 7.4%. A magnified view of the low-angle regions of the XRD results is shown in Figure 7a. Compared with wires, the deposited sample possessed a larger full width at the half-maximum intensity of the selected peak in degree, indicating finer grains and a higher density of dislocations. The density of the dislocations used in Equation (2) was estimated via the Williamson-Hall method, as follows [34]:

$$\beta_{hkl} \cos\theta = \frac{0.94\lambda}{D} + 4\varepsilon \sin\theta \quad (2)$$

where β_{hkl} is the full width at the half-maximum intensity of the peak in degree calculated by the equation:

$$\beta_{hkl} = (\beta^2_{hkl\ \mathrm{M}} - \beta^2_{hkl\ \mathrm{In}})^{0.5} \quad (3)$$

where $\beta_{hkl\ \mathrm{M}}$ and $\beta_{hkl\ \mathrm{In}}$ are the observed and instrumental peak broadening, respectively. D is the average size of the grains. ε is the micro-strain. The fitting curves are presented in Figure 7b. The density of the dislocations for depositions was calculated by [35]:

$$\rho = \frac{2\sqrt{3}(\varepsilon^2)^{0.5}}{Db} \quad (4)$$

where b is the Burger vector taking 0.286 nm for Al-7Mg alloys. Compared with the wires, Al-7Mg alloy depositions possessed a higher density of dislocations. It was attributed to the severe plastic deformation process, during W-FSAM technology. A large amount of dislocations generated and multiplied, estimated to be 1.77×10^{13} m^{-2}, could enhance the yield strength of the depositions.

3.3. Tensile properties

Tensile engineering stress-strain curves of the W-FSAM processed Al-7Mg aluminum alloys are presented in Figure 8a, b. The deposited samples reached a yield strength (YS) of 263 ± 9 MPa, ultimate tensile strength (UTS) of 415 ± 11 MPa, and elongation (El.) of 31.0 ± 2.0% along the travelling direction. The tensile engineering stress-strain curve of rolled AA5083 plates was also presented as a comparison and $\sigma_{0.2}$, UTS, and El. were 127 ± 7 MPa, 353 ± 9 MPa, 23 ± 3%, respectively. The tensile results of the deposition direction for the deposited sample are shown in Figure 8b. The YS, UTS, and El. were 269 ± 2 MPa, 401 ± 3 MPa, and 20.0 ± 2.0%, respectively. The mechanical properties of the deposition direction were close to those along the travelling direction, proving that the mechanical performances were virtually independent of direction. The excellent mechanical properties along the deposition direction were attributed to the sound bonding achieved between the adjacent layers, and the adjacent layers were mixed and joined by the designed rotational pin. Both mechanical performances along the deposition and travelling direction are significantly superior to the commercial 5xxx aluminum alloys, as Figure 8a shows. Actually, the Al-7Mg alloy deposition presented an excellent performance equivalent to commercial 2xxx aluminum alloys [36].

The corresponding fracture location for the deposited sample is also shown in Figure 8a. It was intriguingly found that although the samples possessed excellent ductility, there was no necking found in the test specimens. It was illustrated that the high ductility was attributed to the high strain hardening capacity leading to the large uniform elongation. The high strain hardening capability inhibited localised plastic strain instability and necking [37, 38]. True stress-strain curves of the W-FSAM processed Al-7Mg aluminum alloys are presented in Figure 8c. The Portevin – Le Chatelier (PLC) effect distinguished by irregular stress serrations was found. It was natural to consider the interactions between mobile dislocations and solute clusters that are usually associated with the concept of dynamic strain aging (DSA) [39, 40]. The tendency of the YS, UTS, and El. versus the height of the sample was shown in Figure 8d, indicating that the deposited sample possessed homogeneous mechanical properties. All the tensile properties presented in Figure 8 suggested that the Al-7Mg aluminum alloys fabricated by W-FSAM achieved a good combination of strength and ductility, attributed to grain refinement and the strengthening effects of the Mg solute atoms. Furthermore, the high strain hardening capacity was attributed to the interaction between the supersaturated solute atoms and moving dislocations.

To further describe the enhanced strain hardening capacity, Figure 9a presents a phenomenon of discontinuous yielding behaviour, but it did not appear in the rolled AA5083 aluminum alloys. Lüders strain is an inherent feature of Al-Mg alloys with annealed fine grains, attributed to the DSA effect, as well as the interactions between the mobile Mg solute atoms and dislocations [41]. The yield point elongation is also a common phenomenon in Al-Mg alloys, particularly in fine grain alloys [42]. For the sample presented in Figure 9a, the

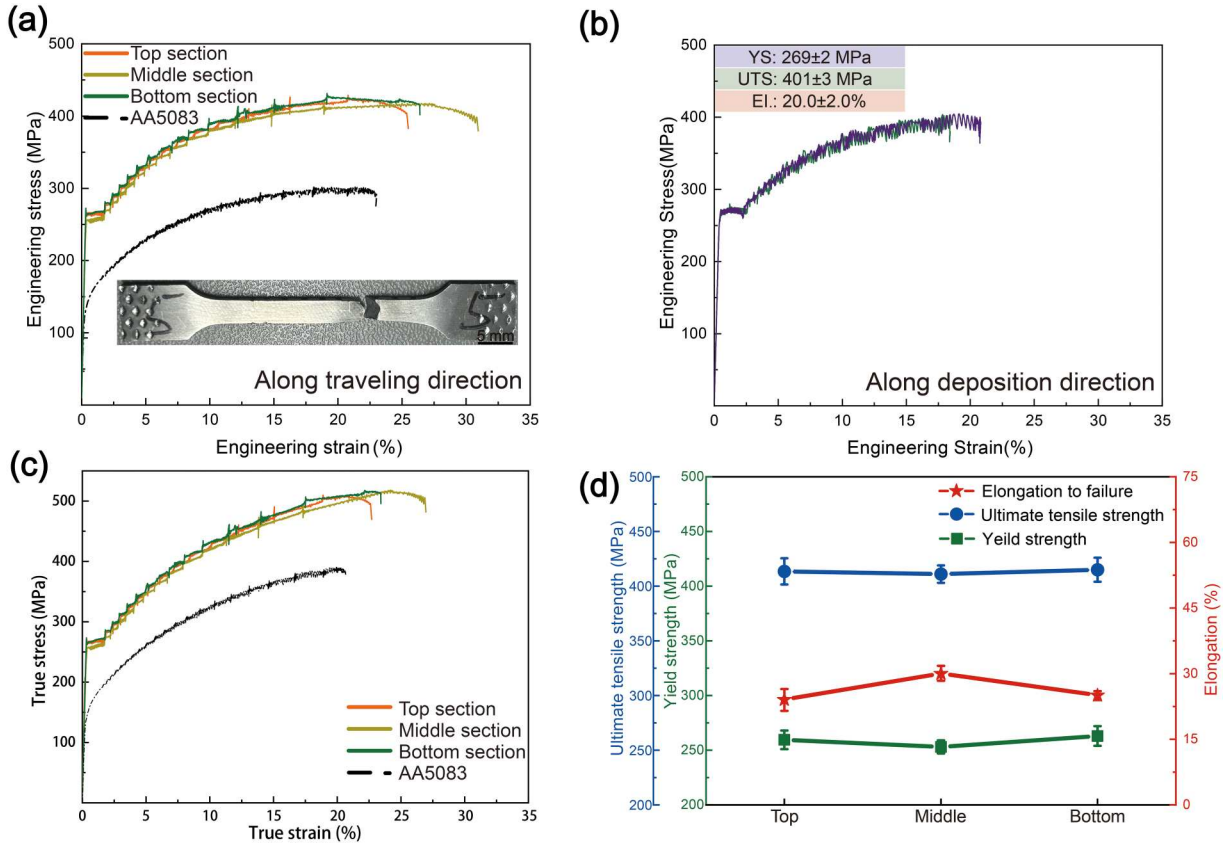


Figure 8. Results of the uniaxial tensile tests: (a) engineering stress-strain curves along the travelling direction of the depositions and the curve of AA5083, (b) engineering stress-strain curve along the deposition direction, (c) true stress-strain curve, and (d) tensile properties versus the height of the depositions.

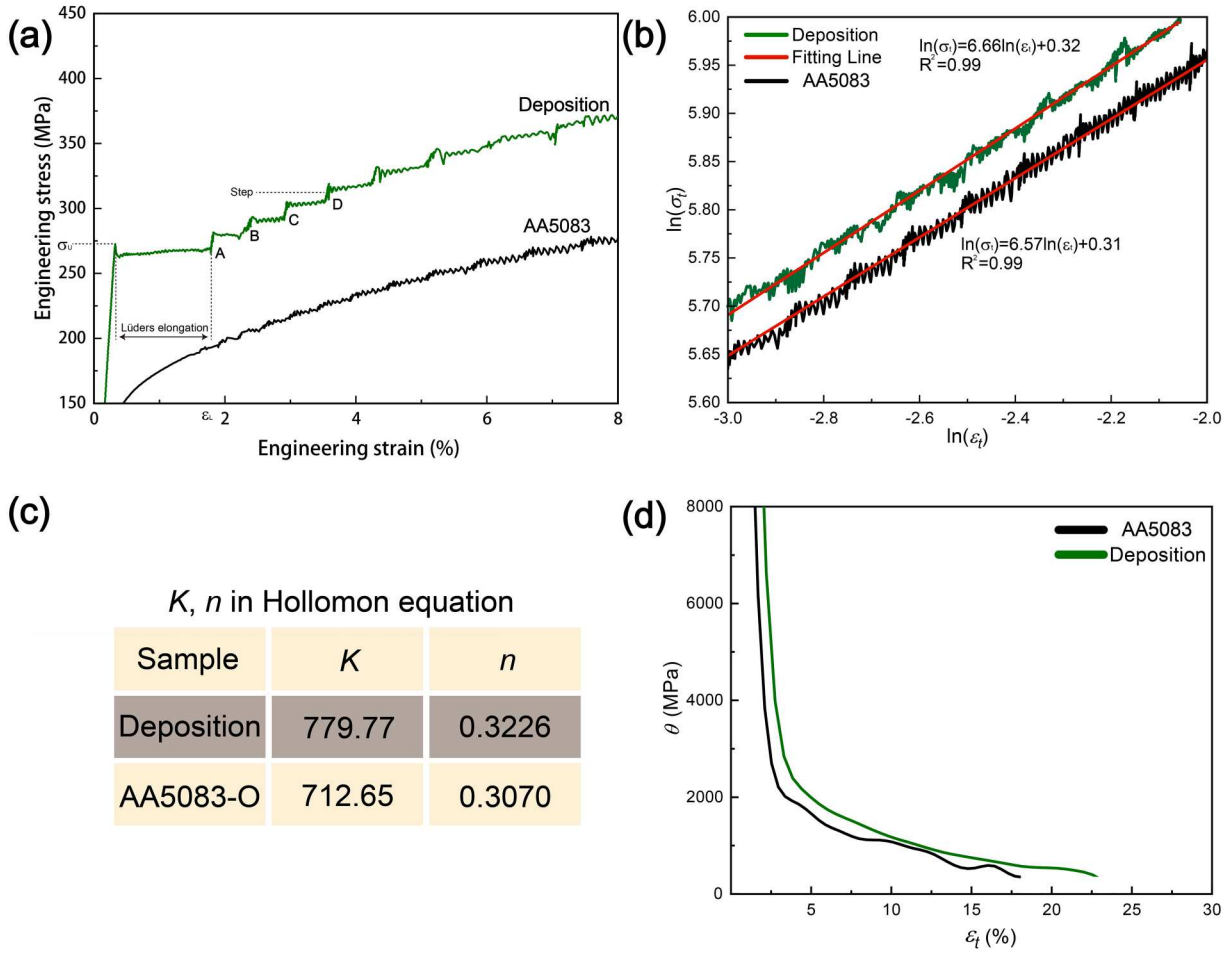
Lüders strain was as large as 1.5%. Currently, scholars generally attribute the discontinuous yielding behaviour to the following reasons: 1) Cottrell atmosphere formed by solute atoms; 2) lack of mobile dislocations due to grain refinement and annealing [43]. During the W-FSAM process, the samples went through dynamic recrystallization, and mobile dislocations were mostly removed, as shown in Figure 6. Besides, solute Mg atoms caused strong dislocation pinning hindering the motion of mobile dislocations. As a result, the yield drop phenomenon appears since higher stress is required to activate dislocations and dislocation sources. Furthermore, in A, B, C, and D four apparent steps were indicated in Figure 9a, and each stress increment between adjacent steps was 15 MPa. That stepped characteristic is different from conventional results. The process of Lüders band propagation was understood as dislocation generation within the band and the introduction of dislocations into adjacent regions a short distance ahead of the band front [44]. Thus, the beginning and end of Lüders strain depended on the resistance of the introduction of dislocations, due to unyielding grains. Some microstructural differences might have caused a discontinuity in the Lüders strain [45]. The magnified curves presented in Figure 9a display the strain-hardening behaviour. The empirical Hollomon relationship was used to approximate the true stress (σ_t) – true strain (ε_t) relationship between the end of Lüders elongation and the UTS shown as Equation (5):

$$\sigma_t = K\varepsilon_t^n \tag{5}$$

where K is the strength coefficient, and n is the strain-hardening exponent. The ln σ_t – ln ε_t curves are presented in Figure 9b. Using the least square method, the K and n can be calculated respectively as listed in Figure 9c. Compared with rolled 5083 plates, the deposited samples owned a larger strain-hardening exponent generally. It enhanced overall strain-hardening capability which contributed to the uniform deformation, resulting in high ductility. The strain-hardening rate θ can be calculated by Equation (6), shown in Figure 9d:

$$\theta = \frac{d\sigma_t}{d\varepsilon_t} \tag{6}$$

It can be seen that θ of the depositions is higher than that of 5083, representing larger θ and superior strain



Sample	K	n
Deposition	779.77	0.3226
AA5083-O	712.65	0.3070

Figure 9. Analyses of strain hardening ability: (a) the discontinuous yielding behaviour, (b) the ln σ_t – ln ε_t curves and the fitting curves, (c) the K, n in Holloman equation and the Lüders true stress (σ_{t-L}), and (d) the θ against the ε_t curves.

hardening capacity. Strain-hardening rate θ decreased steadily against the true strain (ε_t). It enhanced the capacity for depositions of resistance to plastic deformation and achieved a good combination of strength and ductility. In addition, all the tensile curves of depositions showed serrated behaviour caused by DSA, during plastic deformation. It is a common phenomenon in Al-Mg alloys, commonly attributed to that Mg solute atoms restrict dislocation motion and are responsible for the strain rate dependency of serrated yielding [46].

4. Discussion

4.1. Formation mechanisms

The plasticisation and deposition processes are shown in Figure 10. As the pressure in the barrel elevated, the particles in motion ceaselessly contacted and rubbed against the barrel and the tool where the thermo-plasticised films first formed along, as shown in Figure 10a. Subsequently, the thermo-plasticised film was scraped off as the thickness of the film was larger than the gap between the tool and barrel. The thermo-plasticised zone formed towards the screw edge as shown in Figure 10b. Fragmented particles and the thermo-plasticised materials were mixed and compacted at the boundaries of the zone and further deformed even crushed to promote the plasticisation process. The substrate and the former layers were thermo-plasticised and mixed with feedstocks due to the action of the pins under the rotating tool. Sufficient material flow promoted sound bonding between the adjacent layers as shown in Figure 10c. The motions of thermo-plasticised materials along the screw edge are explained in Figure 10d. To illustrate the transport of the thermo-plasticised material, the screw was assumed stationary and the barrel was relatively rotating in the opposite direction. Thermo-plasticised material was dragged by the inner wall of the barrel so that it moved from position A to position A′. The dragging force (F_g) included the friction force and the metallurgical bonding between thermo-plasticised materials and the inner wall of the barrel.

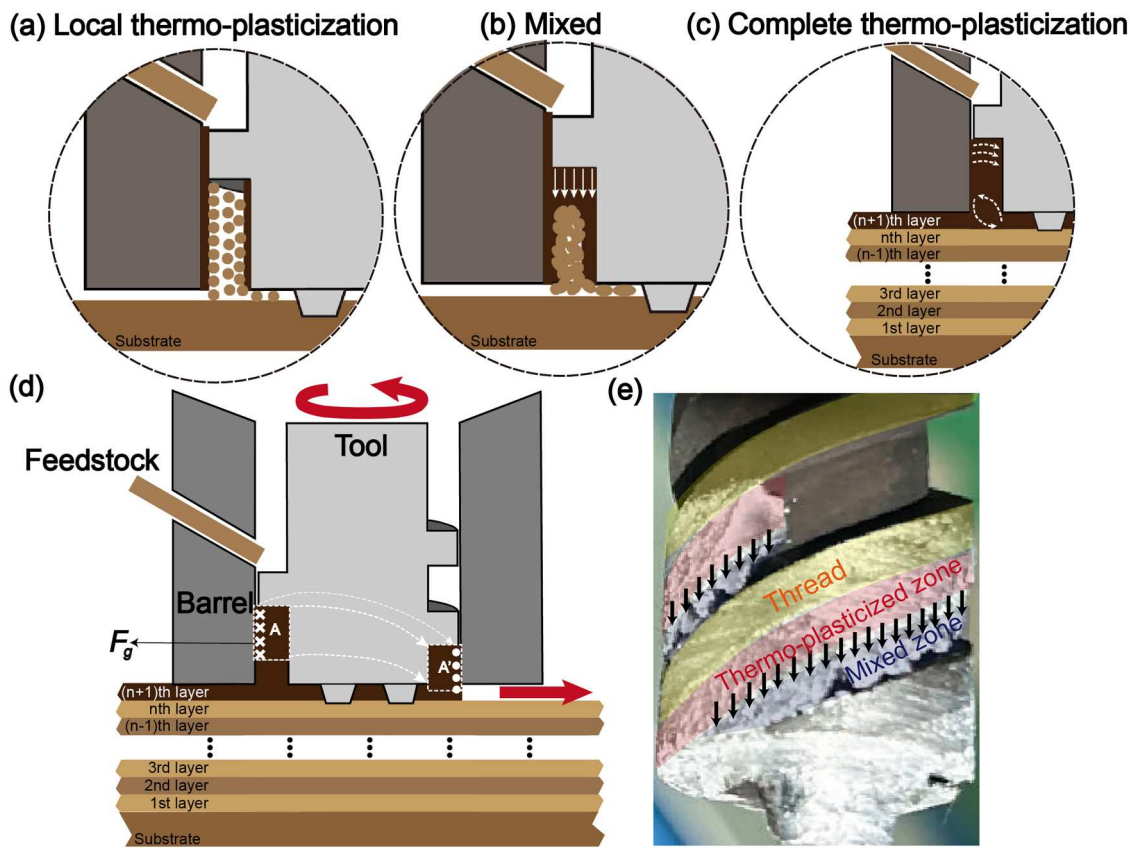


Figure 10. Formation mechanisms: (a) the formation of the thermo-plasticised films, (b) the mixed and compacted procedure, (c) the thermo-plasticisation and deposition process, (d) the transport of thermo-plasticised materials and (e) the actual thermo-plasticised process.

Thus, choosing fitting rotational velocity and making the temperature appropriate for metallurgical bonding to provide the correct F_g, would benefit the transport of materials. Thermo-plasticised materials would be a failure if the temperature was too high to provide the driving force for transport. The picture of the thermo-plasticised process was captured by discontinuing the process as shown in Figure 10e. The plasticised and mixed zone could be observed in the grooves.

4.2. Strengthening mechanisms

As work-hardenable aluminum alloys, the mechanical properties of Al-Mg alloys are mainly determined by grain size, solid solution, and dislocation densities [47]. The high strength of W-FSAM processed Al-Mg alloys was mainly attributed to three mechanisms, i.e. solid solution strengthening by solute Mg atoms ($\Delta\sigma_{ss}$), fine grain strengthening by recrystallization ($\Delta\sigma_{GB}$) and dislocation strengthening ($\Delta\sigma_{dis}$) introduced by elevated-temperature severe plastic deformation as shown in the following equation:

$$\sigma_{yield} = \sigma_0 + \Delta\sigma_{ss} + \Delta\sigma_{GB} + \Delta\sigma_{dis} \tag{7}$$

The interactions between dislocations and solute atoms could reduce the dislocation mobility, resulting in solid-solution strengthening. The YS increment caused by solid solution strengthening ($\Delta\sigma_{ss}$) could be predicted by the following equation:

$$\Delta\sigma_{ss} = HC^n \tag{8}$$

where H and n are material-dependent constants. For Mg solid solutes in commercial Al alloys, $H = 13.8$ MPa wt.%$^{-1}$ and $n = 1.14$ in the case of $\varepsilon = 0.002$ at a strain rate of 6×10^{-3} s^{-1} [48]. Since the yield stress is just a weak function of strain rate in the range between $1 \times 10^{-4} \sim 1 \times 10^{2}$ s^{-1} in Al-Mg alloys [49]. In this paper, the same H and n were used to evaluate the solid solution strengthening from the Mg element. It was calculated that the solid solution strengthening from Mg solute atoms achieved ~138 MPa.

Due to the interactions between dislocations and grain boundaries, YS could be enhanced by the fine grains. The Hall-Petch relation was used to calculate the effect of fine grain strengthening on YS:

$$\Delta\sigma_{GB} = kd^{-\frac{1}{2}} \tag{9}$$

where k is the constant related to strengthening

increment from grain boundaries. For Al alloys, k is usually taken to be 7×10^4 N/m$^{3/2}$ [50]. The fine grain strengthening increment could be estimated by Equation (8), about 58 MPa. The W-FSAM facilitated dynamic recovery and recrystallization of the grains, leading to grain refinement and enhancing the YS in accordance with the Hall-Petch relation.

The density of dislocation density ρ was also a significant value to estimate the strengthening increment from dislocations and to reveal the changing trend of dislocation hardening. The $\Delta\sigma_{dis}$ could be calculated by Taylor relation as shown in Equation (10):

$$\Delta\sigma_{dis} = M\alpha Gb\rho^{1/2} \tag{10}$$

where M is the mean orientation factor for aluminum alloy (3.06), α is the constant for FCC metals (0.2), G is the shear modulus of the alloy Al-7Mg (27.0 GPa), b is the Burgers vector (0.286 nm), and ρ is the dislocation density, and the value was calculated by XRD results and used here. The dislocation strength increases to about 20 MPa.

The strengthening increment from $\Delta\sigma_{ss}$, $\Delta\sigma_{GB}$, and $\Delta\sigma_{dis}$ was add up to about 216 MPa. Taking σ_0 into consideration, σ_{yield} totally reached 226 MPa. Additionally, the precipitates strengthening by Orowan mechanism-($\Delta\sigma_{Oro}$) could also enhance the tensile properties slightly. Thus, the estimated σ_{yield} was close to the experimental value. Ultimately, the Al-Mg fabricated by the W-FSAM process achieved a high yield strength.

4.3. Strain hardening mechanisms

The Mg element could enhance the strain hardening ability determining the uniform elongation and contributing to achieving a good combination of strength and ductility [14]. Figure 11 shows the principle of the high strain hardening capacity. The results and discussion above suggest that the deposited grains are supersaturated with Mg atoms according to the XRD lattice parameter analysis results and the serrated-flow behaviour found in the stress-strain curves. A high concentration of Mg resulted in many micro-regions with

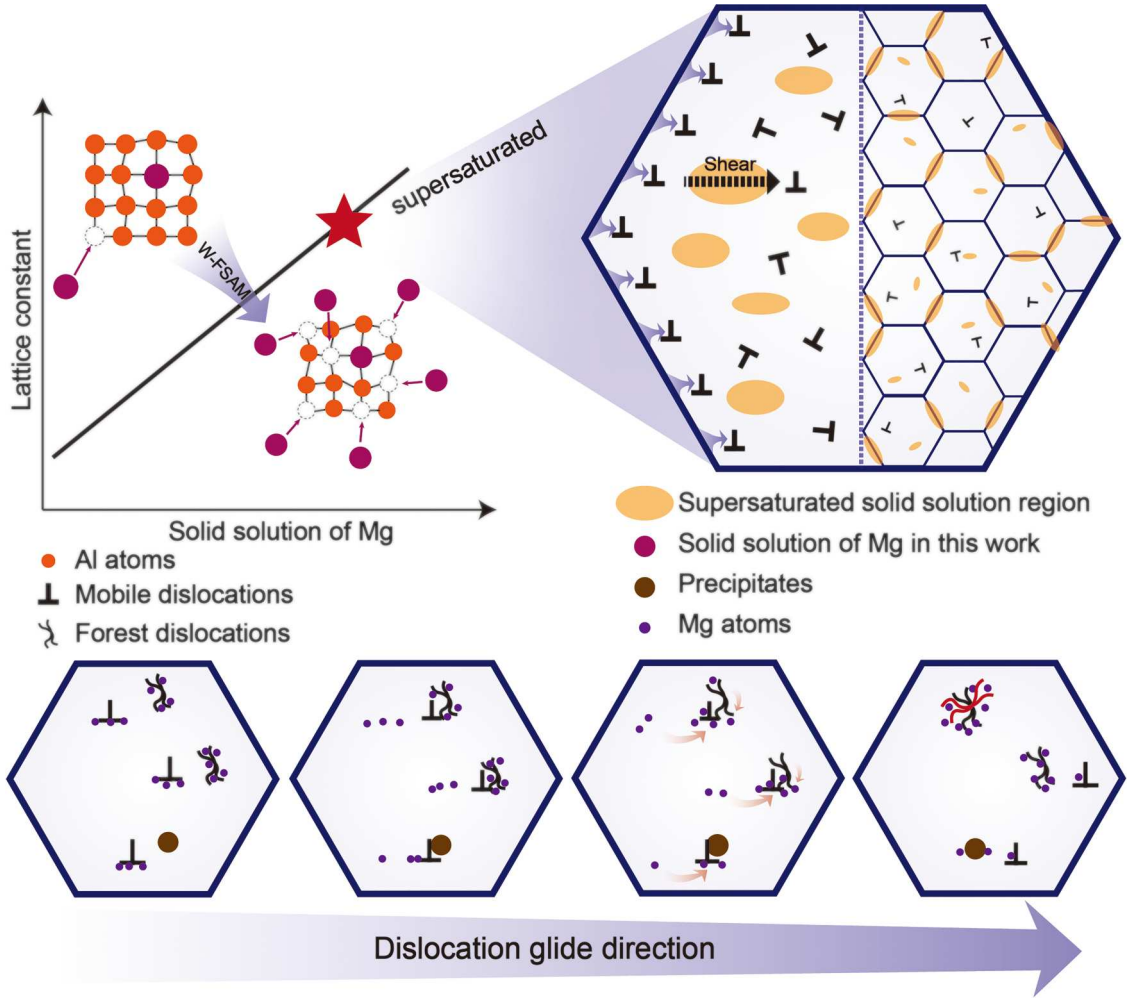


Figure 11. Schematic of high strain capacity mechanisms.

supersaturated solid solutions of Mg (such as clusters, GP zones, coherent precipitates, etc. [33, 51, 52]). Moving dislocations could shear the regions because they were coherent with the matrix. That did not decrease the moving distance but could apply finite resistance on moving dislocations in addition to the usual Peierls stress [53]. Thus, it enhanced the strain hardening capacity of the deposition. The grain boundaries possess a higher solution of Mg atoms due to the severe plastic deformation [54]. The elevated concentration of solutes in grain boundaries can suppress the emission of dislocations from such boundaries due to solute drag [55]. The equiaxed fine grains achieved higher strain hardening capacity than coarse grains via this mechanism.

Furthermore, the interactions between solute atoms and moving dislocations are discussed below. The difference between the atomic radius of the Mg atom (1.71 Å) and the Al atom (1.43 Å) resulted in a strong driving force for the direct transport of solutes from the compressive stress field towards the tensile stress field of dislocations [56]. The solute atoms ultimately formed the Cottrell atmosphere, pinning the forest and mobile dislocations [57]. They were stuck to the solute atoms before dislocations moved, as shown in Figure 11. During the severe plastic deformation process, mobile dislocations escaped the Cottrell atmosphere and glided along the slip plane. When they were hindered by obstacles like forest dislocations, precipitates, and grain boundaries, the previous solute atoms could diffuse and pin them again, which resulted in the DSA effect. The interaction between moving dislocations and solute atoms also enhanced the strain hardening capacity of the deposition. Besides, the forest dislocations acted as obstacles to hindering the motion of the mobile dislocations. The solute atoms located on the forest dislocations could diffuse to the stopped mobile dislocations, which promoted the transition to the category of forest dislocations. Due to the low stacking fault energy of Al-7Mg alloys, this transition made the dislocations unable to leave their primary slip plane and eventually pile up in this plane. Piled-up dislocations led to the strain hardening of the deformed region and the dispersion of the deformation to the neighbouring regions, thereby reducing the degree of strain localisation and promoting spread-out deformation [58]. Thus, the deposited Al-7Mg alloys possessed excellent strain hardening capacity.

The W-FSAM processing Al-7Mg aluminum alloy exhibited superior mechanical properties attributed to the supersaturated solid solution structure and equiaxed fine grains. Avoiding the formation of eutectic Al-Al_3Mg_2

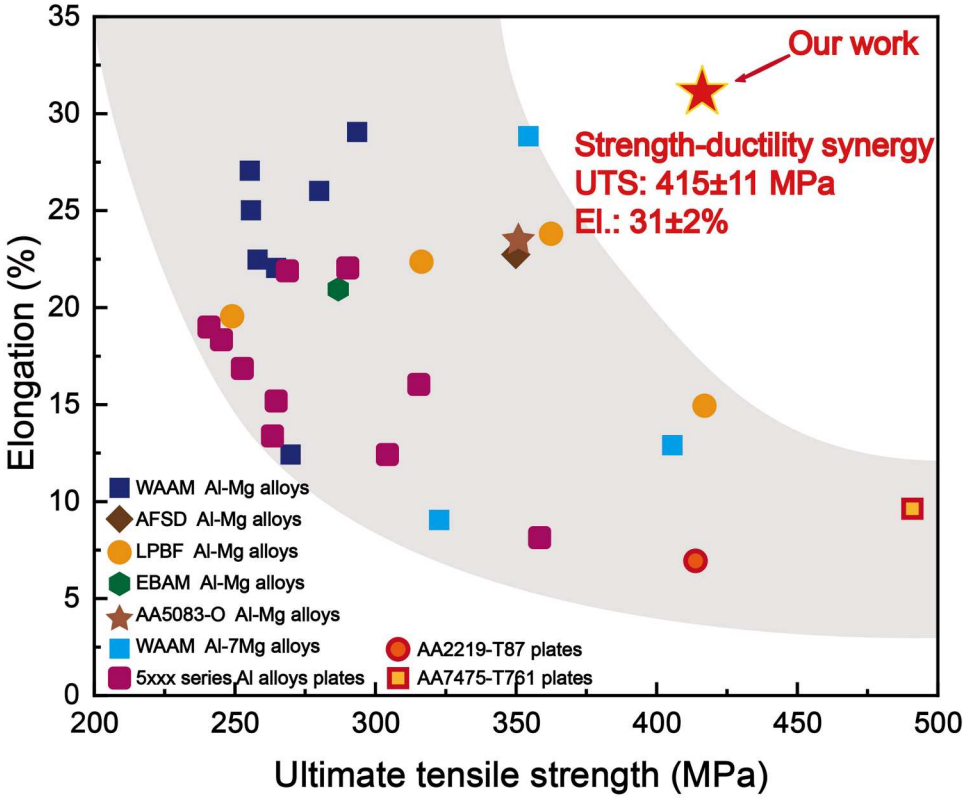


Figure 12. Plot of the tensile performances in relevant literature and our work.

phases along the grain boundaries, W-FSAM fully exploited the enhancement of solid solute Mg atoms. The mechanical properties of kinds of commercial 2xxx, 5xxx, and 7xxx series aluminum alloy sheets such as AA5083, AA2219-T87, and AA7475-T761 as well as the Al-Mg alloy components manufactured by other methods are shown in Figure 12 [28, 36, 47, 59–70]. Contrasting these findings, the Al-7Mg alloy components fabricated by W-FSAM attained an improved mechanical performance, striking a balance of elevated UTS and sparkling elongation. The ultimate tensile strength was equivalent to the AA2219-T87 plates and had a higher ductility. It indicated that Al-7Mg aluminum alloy components could be fabricated with various shapes and structures via W-FSAM and possessed excellent mechanical performance directly, showing potential to be applied in high load-bearing conditions.

5. Conclusions

W-FSAM was proposed and successfully utilised to fabricate Al-7Mg alloy components with homogeneous microstructures and superior, isotropic mechanical performances without further heat treatment. The components were mainly strengthened by refined grains and supersaturated solute atoms. This enhanced the yield strength and improved the work hardening capacity via the interaction between the solution atoms and mobile dislocations. The following conclusions were extracted:

1. Screw and barrel were innovatively designed and used to achieve depositing processes. The sound bonding between adjacent layers was achieved by the stir and forging action from the tools. The components obtained homogeneous microstructure and excellent mechanical properties via this process.
2. Supersaturated solid solution structures were obtained due to the severe plastic deformation, avoiding eutectic Al-Al_3Mg_2 phases along the grain boundaries. The grains were refined with a size of 1.27 ± 0.64 μm due to the dynamic recovery and recrystallization process.
3. Severe plastic deformation introduced from the process resulted in a supersaturated solid-state structure with equiaxed and fine grains, contributing to the yield strength. The high ductility was attributed to the supersaturated solute Mg atom.
4. The YS, UTS, and El. reached 263 ± 9 MPa, 415 ± 11 MPa, and $31.0 \pm 2.0\%$, and reached 269 ± 2 MPa, 401 ± 3 MPa, and $20.0 \pm 2.0\%$ along depositing direction, showing strong-strength-ductility. The components possessed superior mechanical properties to the regular commercial products.
5. W-FSAM for high-magnesium-content aluminum alloys achieved the manufacturing of high-strength aluminum alloy components with near-net-shape, without post-heat treatment. It showed great potential to impetus the application of Al-7Mg alloys in high load-bearing conditions.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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